AEROSENSE

HOME CARE ASSISTANT

Development Guidebook

V2.2

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support@ax-end.com

AxEnd, Inc.
www.ax-end.com
Redondo Beach, california

HISTORY

Date	Version	Modification
04-29-2021	V0.5	Release.
05-25-2021	V1.0	New format with link test tool.
2021-09-06	V1.2	Design change: remove STM32.
03-03-2022	V1.3	Design change: adopt deep learning algorithm.
03-14-2022	V1.4	New format. Edit description of deep learning.
01-25-2023	V2.0	TCP protocol.
07-25-2024	V2.1	Specification.etc
02-21-2025	V2.2	Multi-target detection. Walls for room layout.

AEROSENSE ASSURE

Contents

AEROSENSE ASSURE

Before you start:

Please make sure that you are well prepared.

What do you need?	What it does?
The AeroSense Assure device	A radar that monitors movements.
AeroSense Assure developer package	It contains: **AeroSense Assure and APP user manual** :a manual to install the device. (Please note that for developers, it is recommended to go through this file before register the radar in APP.)
	AeroSense Assure SDK guidebook : the SDK guidance for developers.
	AeroSenseAssure_SDK and sample codes: sample codes for developers.
	AeroSense Assure communication protocol
HomeCareAssistantTool	A software for configuration.
A Windows laptop or PC. (If you are using a Mac, pls install a VMware.)	Used to setup your AeroSense Assure or to connect it to your server.
A 2.4GHz WiFi network	The AeroSense Assure need to be connected to your WiFi network to work properly.

1. Introduction

This manual is intended as a guidance for developers to use the AeroSense Assure and to develop their own codes or functions using the SDK we provide. Please contact our tech support for the SDK and sample codes.

Figure 1 The AeroSense ASSURE: the device

2. Specification

Radar type	FMCW
Frequency	60GHz
Detection range	≥7m (23ft) corner mount ≥5m (16.4ft) wall mount
Horizontal Beam	-45°~45°
Elevation Beam	-45°~45°
Power supply	DC 5V/2A Max
Communication Mode	Bluetooth 5, 2.4G WiFi
Operating temperature	-10 ~ 50 (°C) / 14~122 (°F)
Dimension	148*105*24 (mm) / 5.8*4.1*0.9 (in)

Weight	156g
Communication interface	SDK

Table 1 Specifications

3. Communication Mode

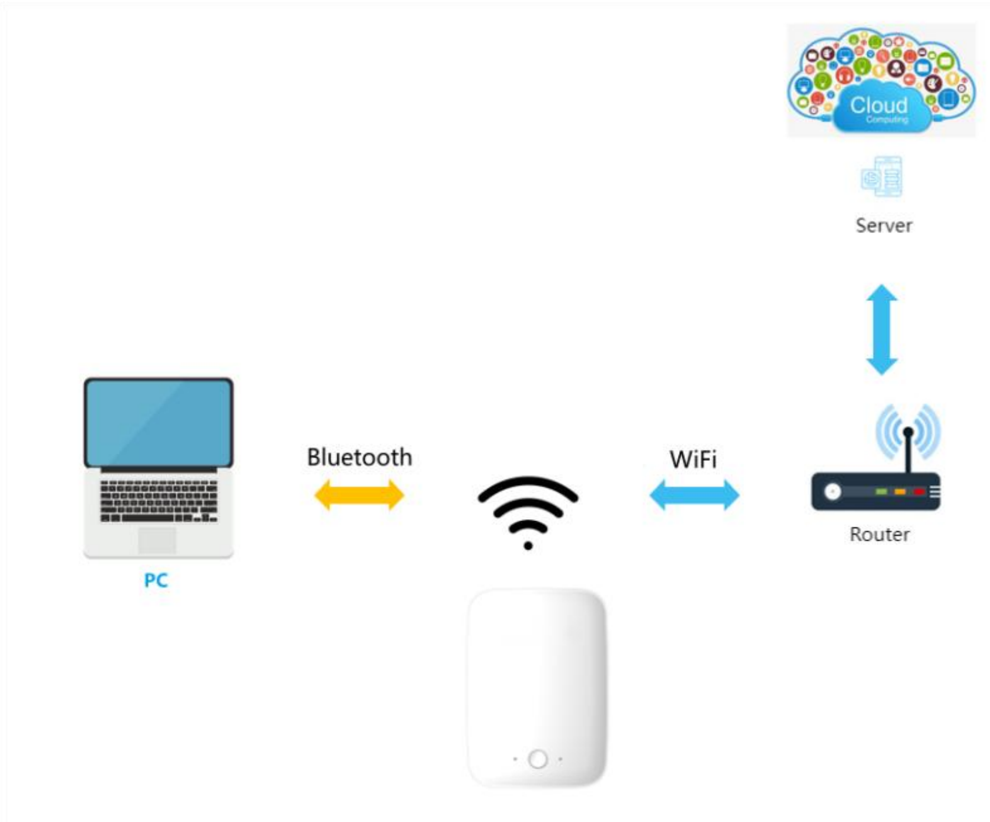


Figure 2 Communication Mode

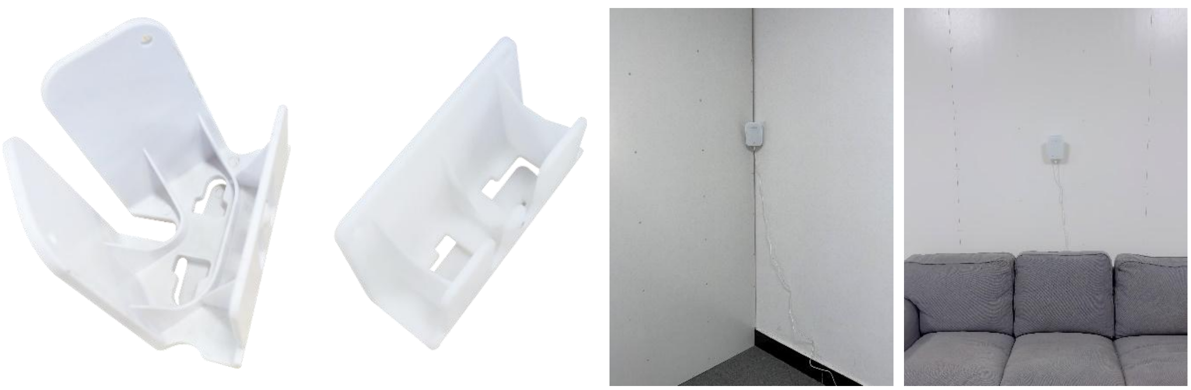
The external interface of the AeroSense is a WIFI protocol. Its WIFI supports AP and STA modes. For firmware 1.8.x.x and older, it supports WiFi@2.4 GHz 802.11b/g/n, and the WEP/WPA/WPA2 security modes. For firmware 2.8.x.x and later versions, Bluetooth 5 is also supported.

4. Device Installation

The device can be either wall mounted or corner mounted, depending on the room layout. Corner mount offers a larger detection range of up to 7m, the recommended setting for living room and bathroom Wall mount is for smaller rooms and can cover a detection range of about 5m, recommended for bedroom especially if the bed is in the middle of the room.

Please note that the wall mount is optional and not included in the standard package. Please ask the manufacturer for more information.

* It is very important to note that you also have to choose the correction mount mode (**Install Type)** in **RoomConfiguration**/**Install Type** as directed in Section 6.6: **0:corner installation** for corner mount, and **1:wall installation** for wall mount.



(a) (b) (c) (d)

Figure 3 The corner mount (a) and the corner installation (c). The wall mount (b) and the wall installation (d)

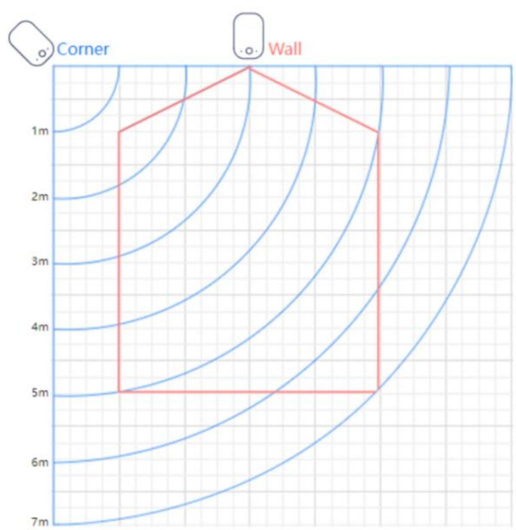


Figure 4 Illustration of the detection range of the wall mount (red) and the corner mount (blue)

5. Connecting to the network

Please plug in your AeroSense Assure. The LED indicator shall flash blue. If not, please press and hold the HOME button for 3-5 sec until it flashes blue. For devices with firmware 1.8.x.x and older versions, please refer to section 5.1, and for devices with firmware 2.8.x.x and later versions, please refer to section 5.2.

5.1. To setup the network in AP mode (firmware 1.8.x.x and older versions).

Make sure your AeroSense is in AP mode. When you get the AeroSense Assure, it should be in AP mode by default (LED flashing blue when plugged in). Otherwise you can press and hold home button for 3-5 seconds until the status indicator flashes blue.

Turn on your computer and connect to the "AS_xxxxxxxxx" network. The suffix is the MAC address of the AeroSense Assure.

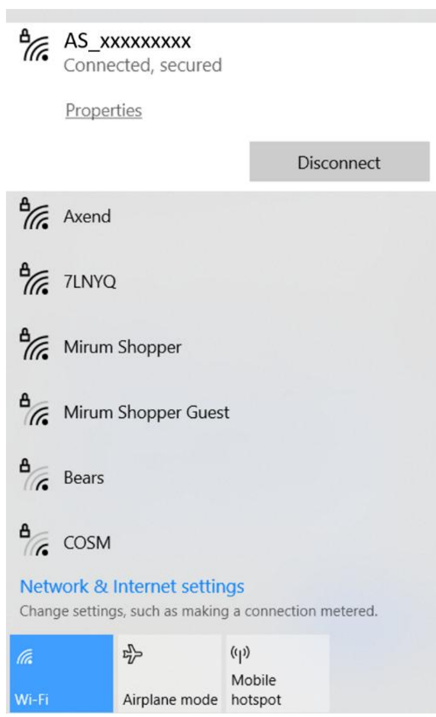


Figure 5 Connecting to AeroSense network (AS_xxxxxxxx) in AP mode

5.2. To setup the network via Bluetooth (firmware 2.8.x.x and later versions).

Please use a computer with bluetooth function. Please do not directly pair your device to your laptop. If it is already paired, please delete the connection.

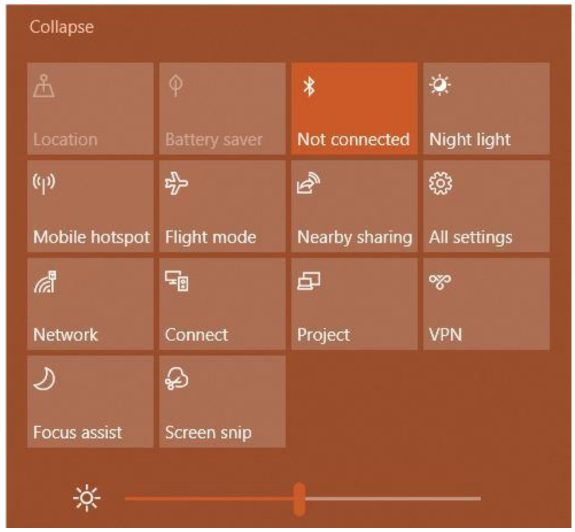


Figure 6 Bluetooth status in your computer

Turn off the firewall of your computer and run the software **HomeCareAssistantTool** to set up the network (SSID, password) and the server (remote server address and port number).

In the panel **config**, click **Scan** in **Step 1** to scan for the bluetooth device (Figure 7-A 1). Click to open the drop down list in Figure 7-A 2 and select the device to be connected.Click **Connect** in in Figure 7-A 3 and wait until the green status bar is 100% completed. If the connection is successful, you will see a green “Connected”. If it fails, please turn off bluetooth in your computer and turn it back on again. Power off the Assure and repeat the step above to connect it to the network via bluetooth.

Enter the **Server IP/Domain**, the **port** number in step 2, and click **Set** to save the changes (Figure 7-B). If the action is successful, you will see a message as seen in Figure 7.

Enter the name and password of the local WIFI and,click **Set** to save the changes(Figure 7-C). If the action is successful, you will see a message as seen in Figure 7.

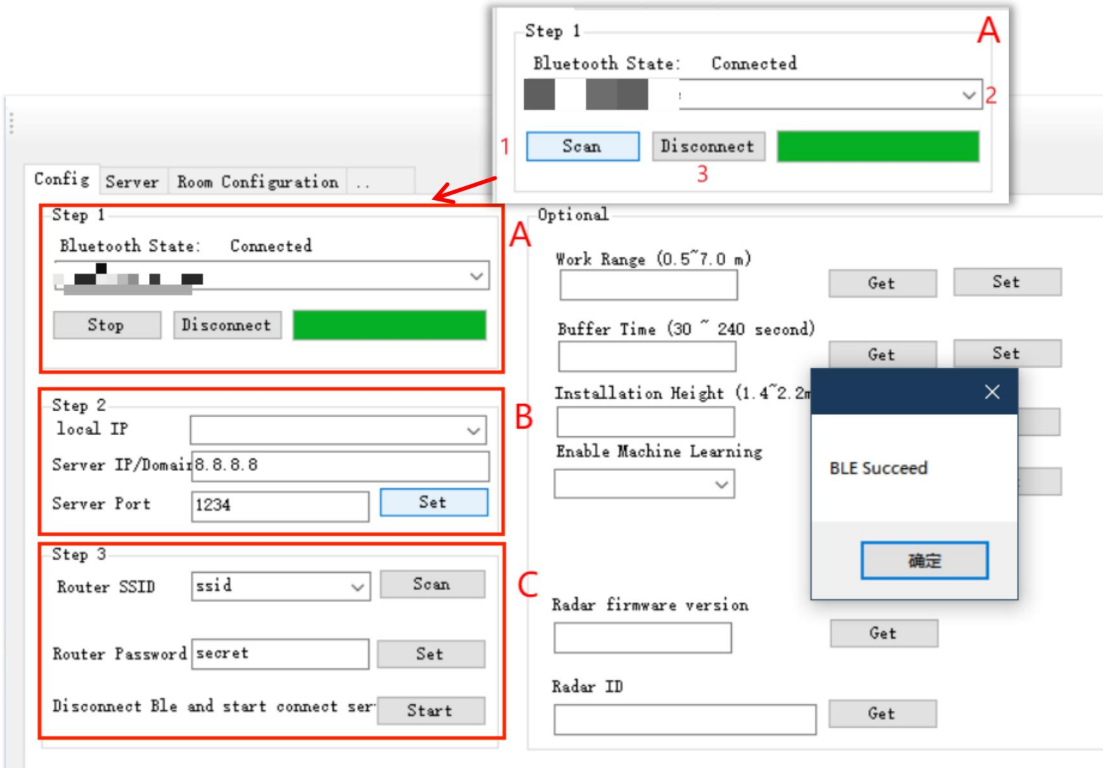


Figure 7 Connect the device and parameter configuration

Click **Start** in **Step 3** (Figure 7-C) to exit network parameter setting. The device will automatically connect to the designated WiFi network and start data transfer via TCP protocol once the network setting is completed. The server will respond to the device via the protocol. The LED indicator will turn from blue flash to constant blue before automatically turned off. If the LED is red, please check if the server is on, and if it is replying device data.

6. The PC Tool and System configuration

Functions of the HomeCareAssistantTool include:

1) The panel **Config** (Figure 8): to set up the network and parameters;
2) The panel **Server** (Figure 9): to setup radar parameters, to check for important system messages, to collect the heatmap, to check room occupancy status, to update or to reboot the device.
3) The panel **Room Configuration** (Figure 10): to manually draw the layout of the room.

If the user is testing the AeroSense Assure using your local server, please note that:

1) Enter the Ipv4 address of the host computer in **Config/Step2/server IP** (Figure 7-B), for example 10.8.2.47. The default port number is 8899.

2) Click **Start Tcp Server** in the panel **Server** (Figure 9-2). Important system information like the device status, firmware version, radar ID will be displayed in **History** (Figure 9-4) and the LED status indicator will be constant blue before automatically turned off.

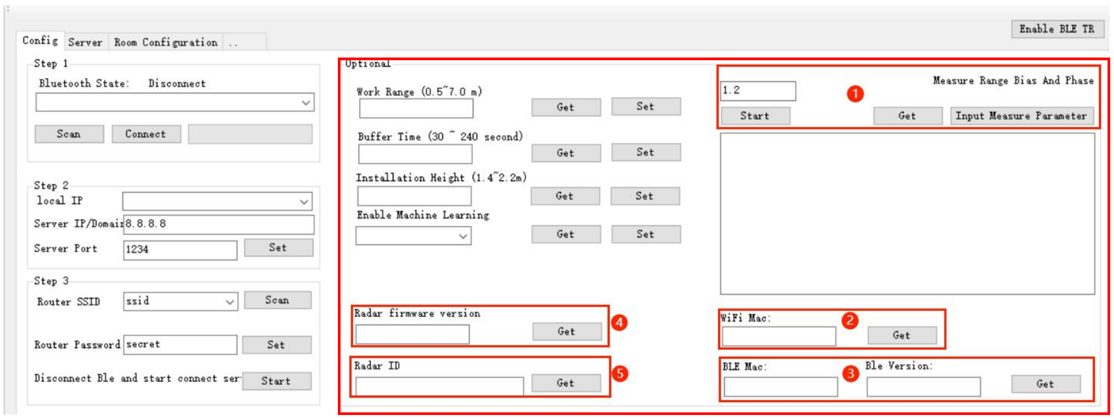


Figure 8 Radar Parameters

1) Note: **Measure Rang/Input Measure Parameter is for factory use only. DO NOT click or change anything.** .
2) WiFi Mac: Click **Get** to get the radar's WiFi MAC address.
3) BLE Mac: Click **Get** to get the radar's Bluetooth MAC address and the firmware version of the communication board.
4) Radar firmware version： Click **Get** to get the current firmware version of the radar.
5) Radar ID: A unique identifier for the radar.

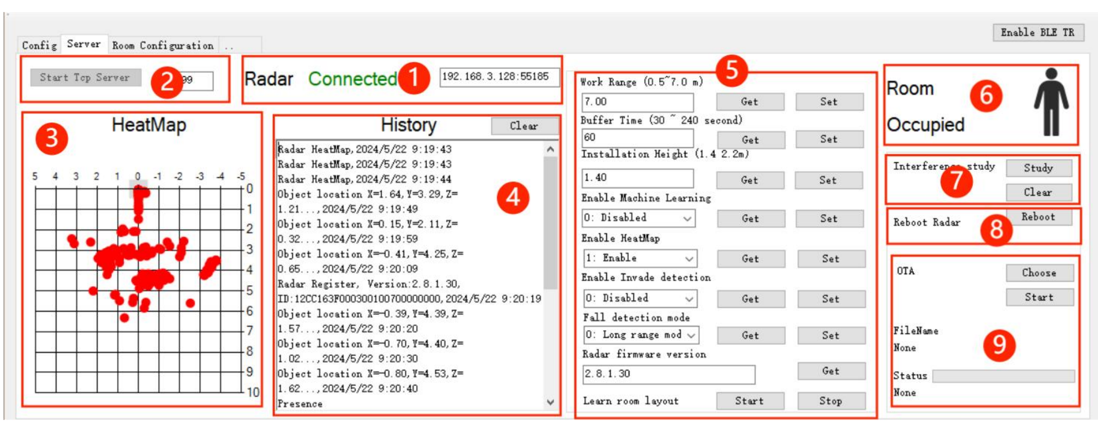


Figure 9 Server navigation

1) Radar status and IP address;

2) Server connection;

3) Heatmap;
4) History: important system messages including device ID and connection status, fall alerts, room occupancy status and heatmap;
5) Radar parameter settings;
6) Room occupancy status;
7) Learning the interference in the room;
8) Reboot;
9) OTA.

6.1. To setup radar parameters

It is necessary to adjust the radar parameters according to the application scenario for the best performance. The user can adjust these parameters Server (Figure 9-5). Simply adjust the value and click **Set** for every parameter that is entered to save the changes.

The radar parameters include:

1) **Work Range**. It refers to the effective coverage of the radar. It should be a figure between 1.0-7.0m with a default set of 7m. It is recommended to use smaller figures for smaller rooms for better effect. For example, if the radar is installed in a small bathroom about 3m in length, please do not use 7m as the work range otherwise the radar may occasional see beyond the entrance.
2) **Buffer Time**. It is the time range between the radar detects a potential fall and sends out an alert. It is a figure between 30-240s with a default set of 60s. During this peroid, if the radar detects any other movements that are large enough to be considered as someone else entering the FOV of the user sits up, the alert will be cancelled.
3) **Installation Height**. It is how high you've installed the AeroSense Assure. Please enter the exact figure (unit: meter). The default height is 1.4m and it should be a figure between1.4 - 2.2 meters, to ensure optimal monitoring results.
4) **Enable Machine Learning**. The Assure adopts a complicated algorithm to learn and remember potential false alerts for better user experience. It is by default off (0: disable, 1: Enable). If the user simply want to test the accuracy for fall detection, please turn off machine learning. If the user want to test the long-term user experience and false alerts, please turn it on.

5) **Enable Heatmap**. It is by default enabled (0: disable, 1: Enable). If it is on, the coordinates or any activities picked up by the device will show up in Figure 9-3, refreshed every minute.

6) **Enable Invade detection**. It is by default disabled (0: disable, 1: Enable). If it is on, the device will send out and invasion alert whenever there is a moving target within the FOV.

7) **Fall detection mode**. You can adjust the sensitivity of fall alerts.

- 0: High alert rate mode. It has higher sensitivity to detect falls, however it may lead to higher falser alerts rates as well. It is recommended for scenarios with higher risk for falls.
    - 1: Low false alert mode. The false alert rate is lower when set in this mode. However, the detection rate may be lower for certain falls as well. .

8) **Radar firmware version**. The user can click to get the current firmware version of the radar.

9) **Learn room layout**. Still under development. Please DO NOT click it.

6.2. To collect the heatmap

If the HeatMap is enabled as described in Section 6.1, the coordinates or any activities picked up by the device will show up in Figure 9-3, refreshed every minute. This is to help the user to better understand the activity pattern and levels within the FOV and for users to develop functions via SDK or communication protocol.

6.3. To check room occupancy status.

The user can monitor the real-time room occupancy, to see if the room is occupied or empty.

6.4. To learn the interference in the room.

Interferences, or so-called noises, in the room can sometimes confuse the radar and the radar may mistakenly consider an empty room as occupied. Common interferences in the room include: fans and ventilations, curtains, fish tanks, and large plantations, or anything that vibrate or rotate. The Assure is capable to memorize fixed interference for better user experience. To do it, make sure that all the know interference in the room is turned on to its maximum setting and there are no other moving targets (persons, pets…). Click **Study** in Figure 9-7 to start the noise collection. The learning takes a minute and will be automatically off when it is over. The status can be read in **History**, as seen in Figure 10. If the user wants to erase data that was learned before, please click **clear**.

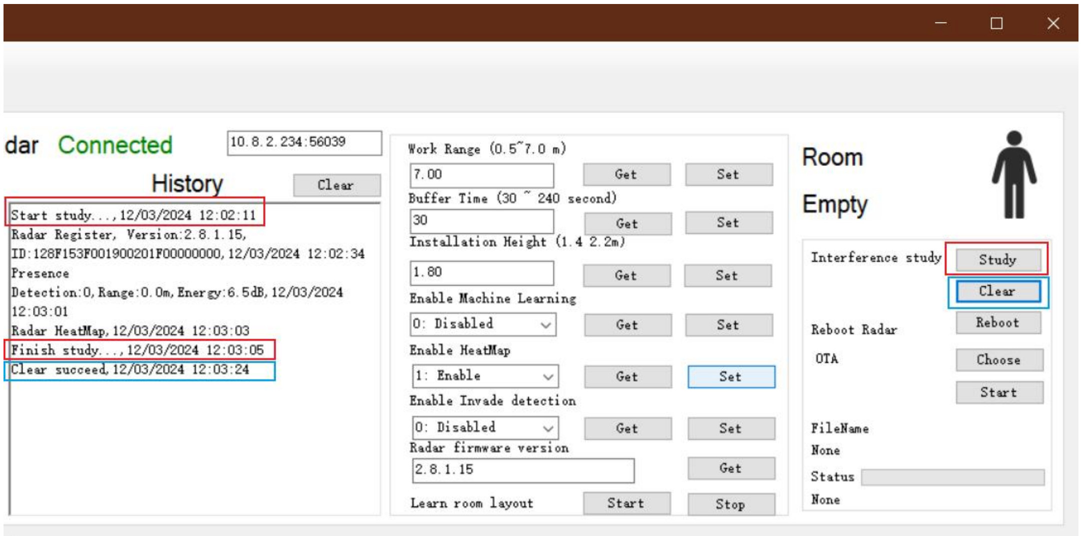


Figure 10 Interference study

6.5. Device reboot

The user can click to Reboot the radar to apply the changes or to restart the system (Figure 9-8).

6.6. To draw the room layout manually

If the user is not happy with the room layout learned by the AeroSense cloud, it is recommended to manually set it up in the panel **Room Configuration**. At the moment, this function is only available for wall-mount devices, which means the user must choose **1** in **Install Type** Figure 11-3.

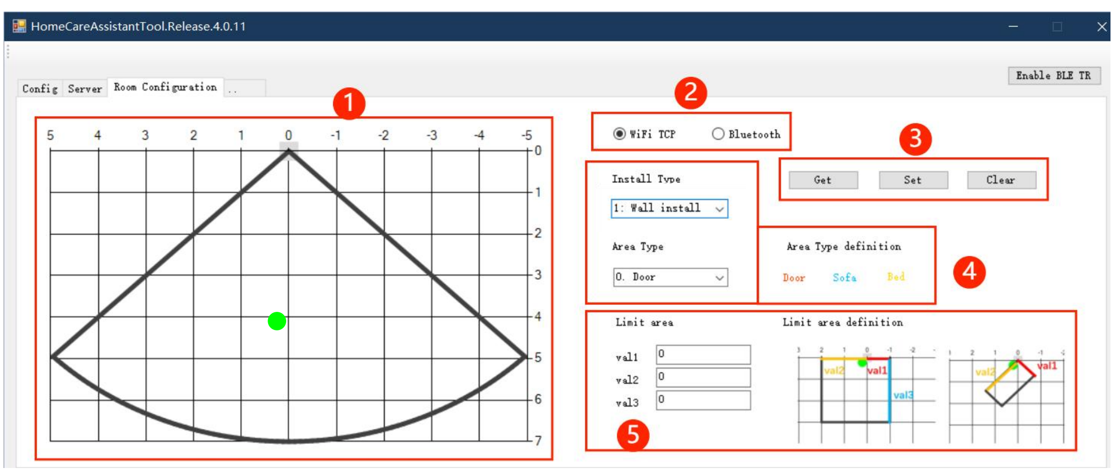


Figure 11 To draw the room layout manually

1) Room layout(Figure 11 - 1). A grid coordinate system is used to demonstrate the room layout. The radar is installed at (0, 0). The area within the think black lines are the detection range. The moving target is displayed as the green dot, refreshed every 10 sec;

2) To select the connect network mode(Figure 11 - 2). Please click either **WiFi TCP** or **Bluetooth** depending on how the device is actually connected to the network;

3) The user can read the area already drawn in the room (**Get**), save the area that is just drawn (**Set**), or to clear the areas that are previously saved in the memory (**Clear**).(See Figure 11 - 3)

4) **Install Type** and **Area Type**(Figure 11 - 4). **Install Type** refers to whether the radar is **wall mount (1)** or **corner mount (0)**, and **Area Type** refers to whether the area selected is the **entrance/exit (0)**, or a **sofa (1)** or a **bed (2)**. Select the correct area type before you draw the corresponding area in the Room layout (Figure 11-1). It is recommended to draw areas based on the real-time display of the green dots for best accuracy;

5) To draw a wall in the room as the boundary (**Limite area** in Figure 11 - 5). Signals beyond the wall will be ignored. It is recommended to use the fall to filter certain fixed interference caused by curtains or reflections, and to limit the detection from within the room.

7. Communicating between the AeroSense Assure and the server

7.1. Introduction

Once the AeroSense Assure has been connected to the server, it will act as a TCP client to communicate with the server. Please refer to the document **Home Care Assistant_Tcp** for the communication protocol.

Proactive register. The device will proactively send the registration request to the server once it is correctly connected to the network.

Server setup. The user can also setup the device parameters in the server.

Example of the data stream is shown in Figure 13. For example, in the message for radar registration, the data format consists of a header and data. The header is composed of Magic, Ver, Type, Cmd, ReqID, TimeOut, Content Len, and Function Code.

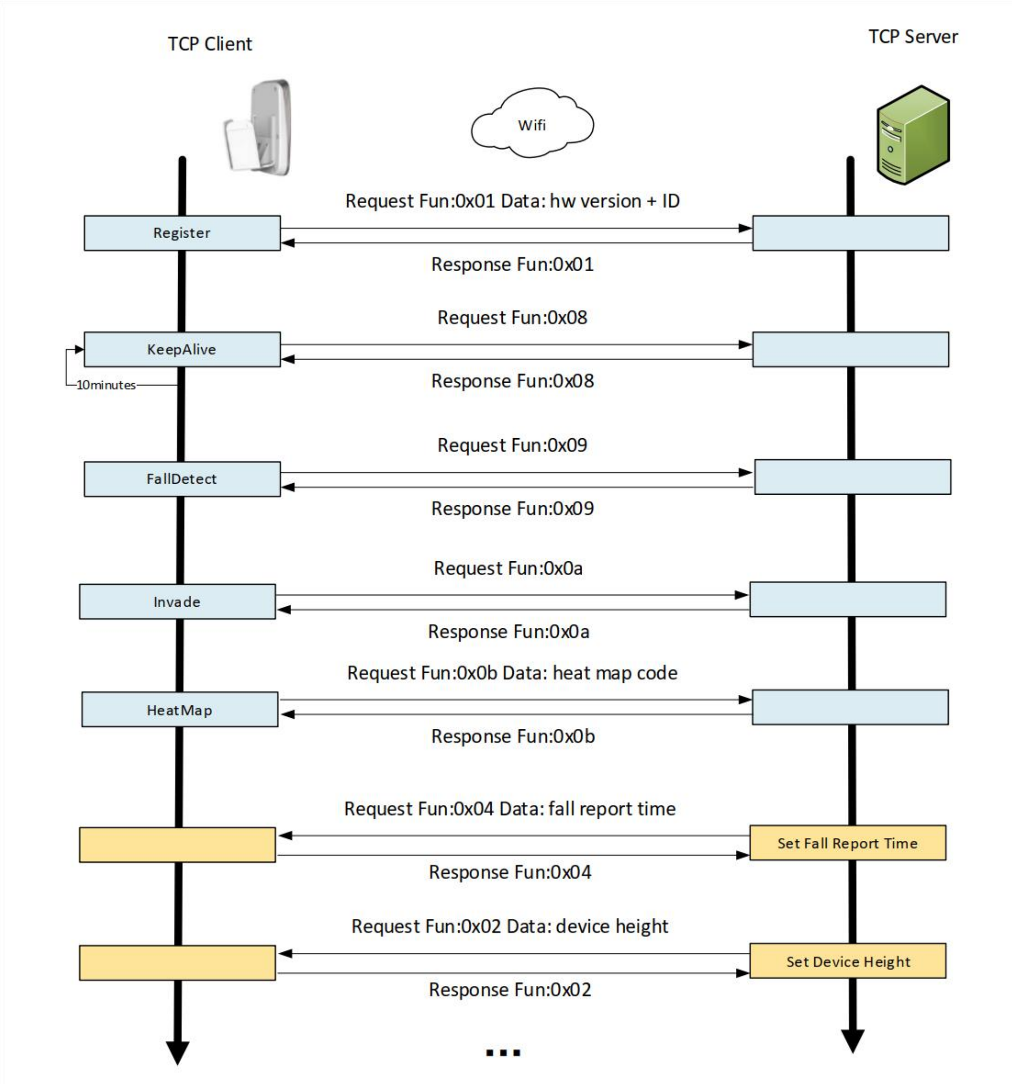


Figure 12 The communication flow between the AeroSense Assure and the server

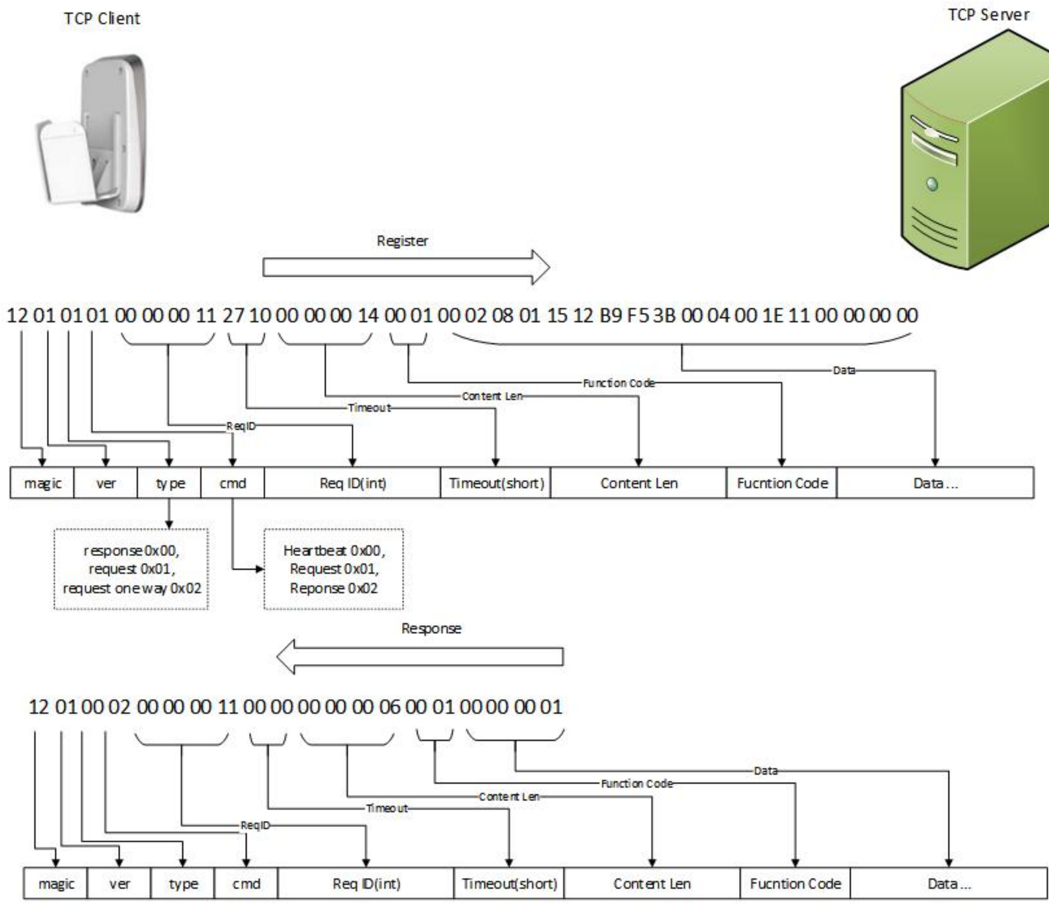


Figure 13 An example of the data stream

- **Magic:** fixed at 0x12 to represent the device.
- **Ver** represents the protocol version (0x01).
- **Type** refers to the message type. It can be a response, a request, and an one-time request without the need of a response).
- **Cmd** is the message command. It can be the heartbeat packet, a request command, and a response command.
- **ReqID** is the request ID, It starts from 0 and is increased by 1 for every time the radar is restarted.
- **Timeout** is the timeout duration for the server.
- **Content Len** is the length of Data plus a Function Code of 2 bytes.

- **Function Code** is the functional byte of this message.
- **Data** is the data bytes.

7.2. Heatmap and encoding

The heatmap represents the activities that are detected by the AeroSense Assure within the FOV over a period of time. The heatmap codes are compressed.

If a valid target is detected, the reported codes will be consisted of three bytes, and in the format of {X, Y, N}, the **coordinate code** in Table 2 . X and Y are the coordinates of the target in the radar coordinate system as shown in Figure 14. N is the number of point clouds of the target.

Code name	Description	Format
Coordinate code	The radar detects the target and outputs the coordinates	{X,Y,N}
Frame number code	The radar does not detect the target and outputs the number of frames that have not been collected	{ the number of frames without valid targets within a certain period of time, 0, 0}

Table 2 The encoding format of the heatmap

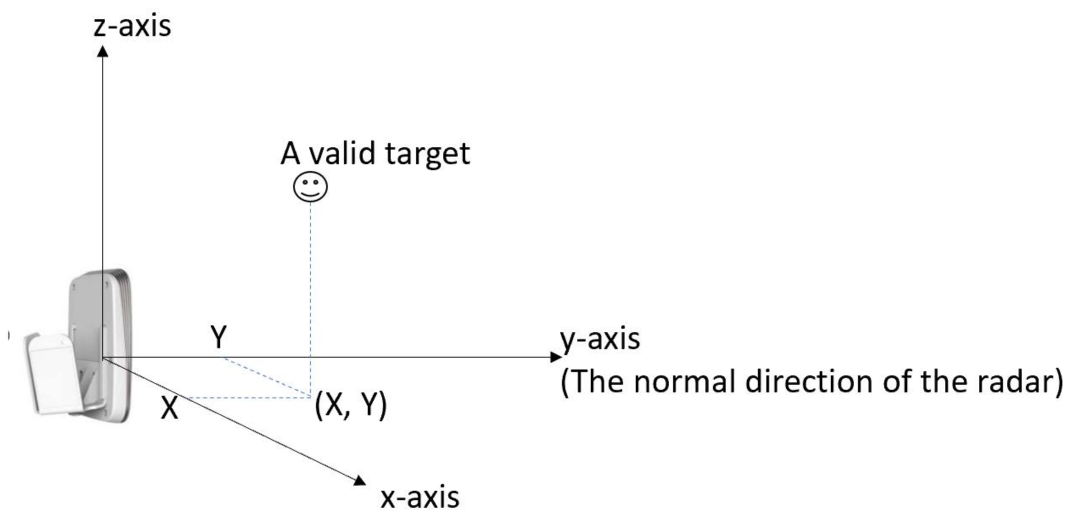


Figure 14 The radar coordinate system.
The radar is (0, 0, 0). The y-axis is the normal direction of the radar.

If there is no valid target, the codes will be in the format of {the number of frames without valid targets within a certain period of time, 0, 0}, the **frame number code** in Table 3. For example, {130, 0, 0} means there are no targets within the 130 frames of data. This format can greatly decrease the data amount. The range of X, Y and N is shown in Table 3-2. It is also worth mentioning that X and Y are in bytes and 1 byte equals to 4cm.

	Byte range	Actual distance range	Remarks
X	-128~128	-510 cm~510 cm	1 byte = 4 cm
Y	0~255	0 cm~1020 cm	1 byte = 4 cm
N	0~255	0~255	number of point clouds

Table 3 Byte range of X, Y and N

7.3. Machine Learning

If you are a developer, it is more than likely that your device is using the firmware without the machine learning algorithm, which is designed to minimize false alerts in everyday use. This is for developers to better test the accuracy of the radar to detect falls. Once you've done the accuracy test, it is recommended to update to the firmware with learning algorithm through the AeroSense Care APP or using the AeroSenseTool for a more thorough experience of the device. Once the learning algorithm is activated, the device will be silent for 48 hours after installation for the radar to learn the room layout. It will take another 7 days for the radar to be more familiar with the activities happening within its FOV to complete the learning process. The learning algorithm is to minimize false alerts for fall detection during day to day living to increase accuracy and reliability of the device.

Once the deep learning algorithm for fall detection is enabled, users will receive two types of fall alerts:

1) **FallDetect**. This is an actual fall alert. The user needs to attend to it.
2) **FallDetect_Neg**. The potential falls picked up by the radar can be judged as false alerts by the learning algorithm. In this case, the user will get a FallDetect_Neg post indicating a false alert.

Time after connecting to the network	Description

0~1hour	Within the first hour, deep learning algorithms will not be activated, so that the user can test the accuracy to make sure that the device is correctly installed. The user will only get “FallDetect” type of data.
1~48hour	The second to the 48th hour after connection, the radar will be silent for the learning algorithm to collect all the possible false alerts, and adapt to the user's behavior patterns and to learn the room layout. It is the so called learning phase. During this time frame, the user will only get "FallDetect_Neg" type of data which **actually means that all falls detected are identified as false alarms .** **Please note that you will not receive any real fall alerts during the learning period.**
After 48hour	After the 48-hour learning period, the device will act as it is intended to. At this point, with the deep learning algorithms fully enabled, the device can distinguish between real fall events and false alarms based on the learned patterns, accordingly reporting "FallDetect" and "FallDetect_Neg." "FallDetect": Indicates that the device has detected a fall and determined it to be a real fall event requiring an alert. "FallDetect_Neg": Indicates that the device has detected a fall but the algorithm has identified it as a false alarm, therefore no alert will be triggered.

Table 4 What happens if the learning algorithm is activated?

Note: The logic of deep learning algorithms aims to balance between the accuracy and false alarm rate of fall detection to maximize user experience. However, in certain scenarios, it may lead to lower sensitivity. Therefore, when demonstrating the device to their customers, it is recommended to disable learning algorithm.

8. Firmware Upgrade

The user can either upgrade the firmware via OTA using either HomeCareAssistantTool, or by the server.

8.1. OTA update by the Server

An OTA update can be done by the server proactively. The update is via segment transfer and CRC verification is introduced for accuracy check, as seen in Figure 15. The firmware will be sent in segments. The length of the content to be sent is (1000, 1000, …N), and the last N bytes are the remaining length. The firmware content is validated by CRC16 modbus.

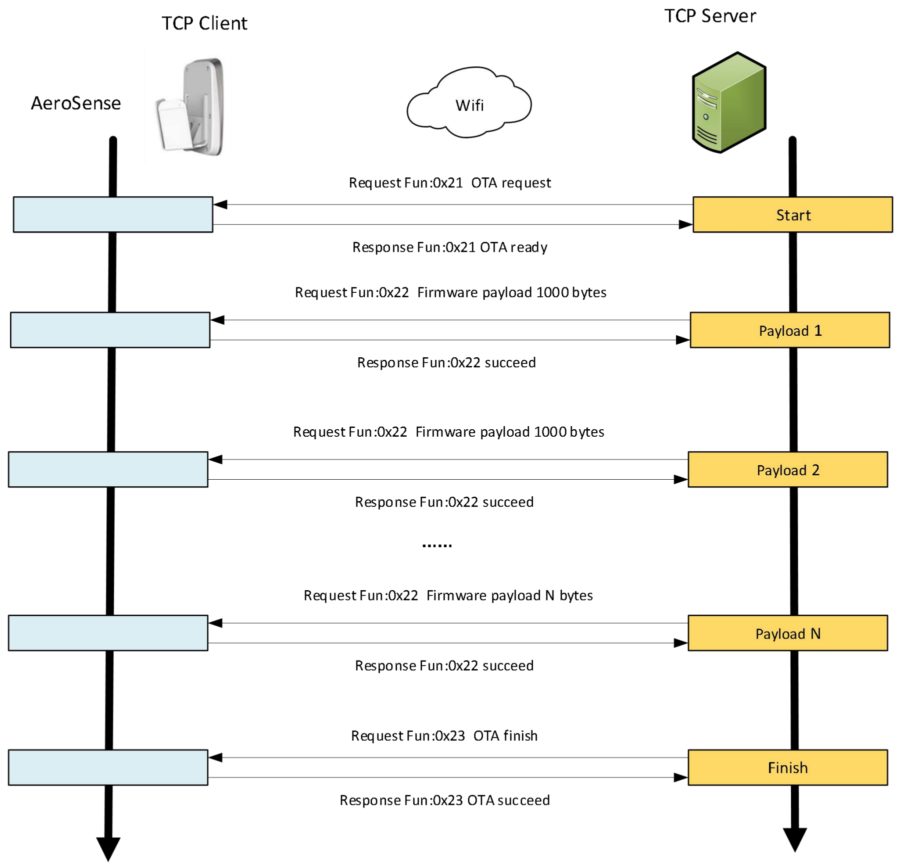


Figure 15 OTA update

8.2. OTA update via FallDectionSensorTool

The user can also use the HomeCareAssistantTool to do the update. It includes device connection, server initiation and firmware selection.

Press and hold the Home button on the device for 3-5 seconds until the radar starts blinking with a blue light. Turn on Bluetooth of the computer.

1) Click to run HomeCareAssistant tool.exe in your computer.
2) In **Config**, scan and connect to the device.
3) In **Config**, setup the serverIP (same as localIP) and port (8899).
4) In **Config**, setup the router's WiFi name and password (2.4Ghz).
5) In **Config**, click **start** to connect the device to the server. See Section 5.2 for more detailed steps.
6) In **Server**, click **Start** TCP Server and wait until the connection succeeds (Figure 11-2).

7) In **Server/OTA**, click **Choose** in Figure 11-9 to select the firmware file. It should be a bin file and name of the file is FallDetectionSensor.Release.2.X.X.X.bin. Click **Start** to initiate the upgrade. Wait until the **Upgrade Succeed** message shows up. If the upgrade fails, simply click **Start** to update again.

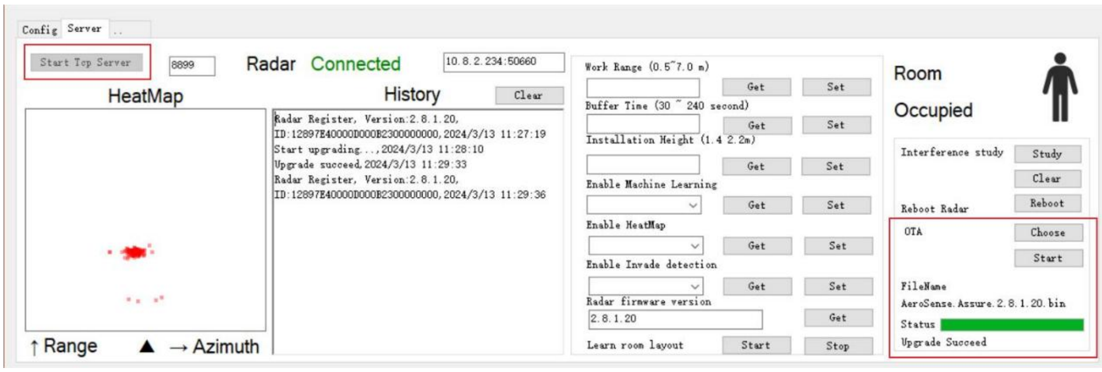


Figure 16 OTA upgrade

8.3. The OTA update of the firmware for the communication board

IMPORTANT! Please note that the firmware of the communication board is different from that of the radar. It is very important to use the correct file. Otherwise, you may loose the device completely.

The name of the firmware for the communication board is :AeroSense_Comm_OTA.Release.x.x.x.x.bin

Click the **Ble Version** button in the lower right corner of the **Config** page (Figure 8) to get the communication board firmware version. If you need to upgrade the communication board, please follow the steps below:

1) Please write down the Tcp Client IP displayed in the upper right corner of the HomeCareAssistantTool.exe PC software (e.g., 10.8.2.34), as shown in Figure 17.

2) Enter the Tcp Client IP (e.g., 10.8.2.34, see Figure 18) in your web browser. Enter the username and password. (For versions 0.0.4.4 and later, both the username and password are admin; for earlier versions, please contact technical support.)

3) Select the correct firmware file (.bin) and click **Upgrade**. Wait until the update is completed(Figure 19). Please note that it is recommended to wait for 5 more seconds even if it shows 100%(For versions 1.0.2.5 later,it shows OK,see Figure 19 - right) to ensure all the data transfer is completed.

Note:

- **Make sure to select the correct communication board firmware. It should have a name of AeroSense_Comm_OTA.Release.x.x.x.x.bin.**
- Firmware versions below 1.0.2.5 cannot be upgraded to TLS encryption versions.

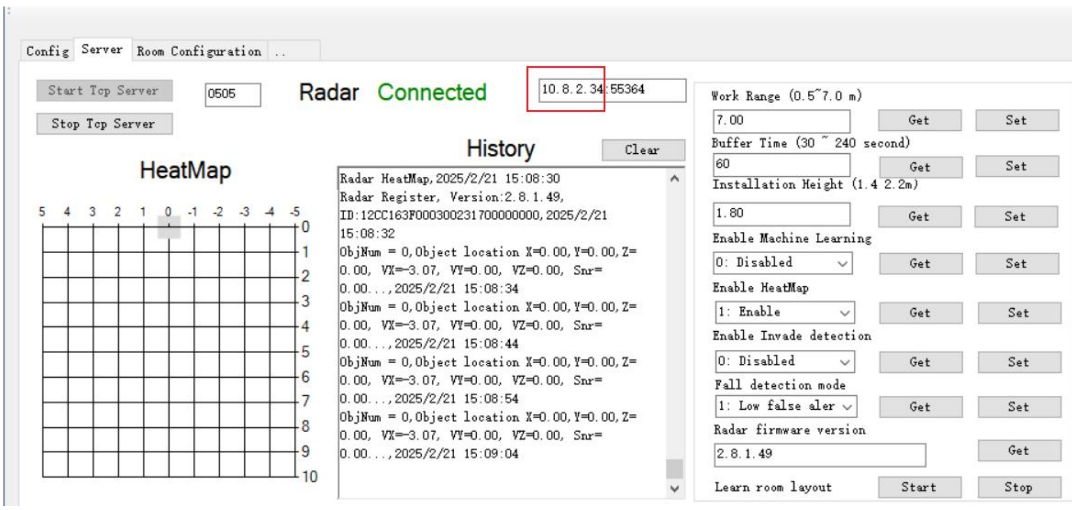


Figure 17 TCP Client IP

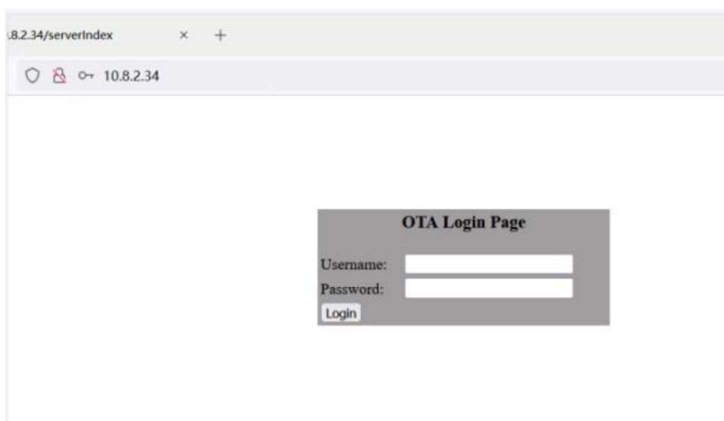


Figure 18 Enter the username and password.

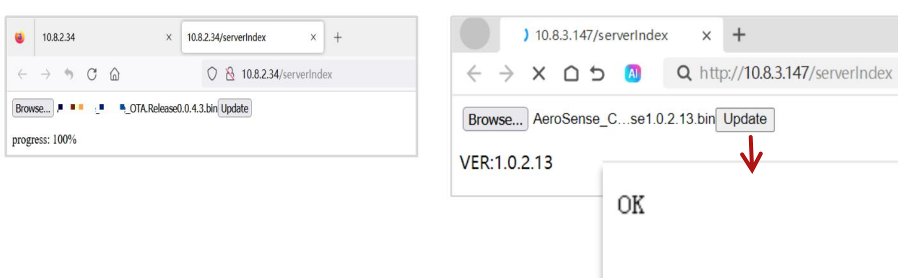


Figure 19 Complete the OTA upgrade for the communication board.